Department of Mathematics, IIT Madras

MA 2102 - Differential Equations, First Semester, 2026-27

Assignment-4: Series Solution at Ordinary Points and Frobenius Method

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1. Find the power series solution of the following differential equations at the indicated ordinary points:

   (a) $y'' + xy' + (x^2 + 2)y = 0$ at $x = 0$;

   (b) $(1 - x^2)y'' - 2xy' + p(p + 1)y = 0$ at $x = 0$, where p is an integer;

   (c) $(1 + x^2)y'' + 2xy' - 2y = 0$ at $x = 0$;

   (d) $2y'' + (1 - x)y' - y = 0$ at $x = 0$;

   (e) $y'' + xy' + (3x + 2)y = 0$ at $x = 0$;

   (f) $(x + 3)y'' + (x + 2)y' + y = 0$ at $x = 0$;

   (g) $xy'' + y' + 2y = 0$ at $x = 1$;

   (h) $y'' + y' - xy = 0$ at $x = 0$;

   (i) $(x^2 + 1)y'' + xy' + 2xy = 0$ at $x = 0$;

   (j) $y'' + (x^2 + 2x + 1)y' - (4 + 4x)y = 0$ at $x = -1$.

2. Find the power series solution of the following initial value problems:

   (a) $(x^2 - 1)\dfrac{d^2y}{dx^2} + 3x\dfrac{dy}{dx} + xy = 0, \quad y(0) = 4, \quad y'(0) = 6;$

   (b) $\dfrac{d^2y}{dx^2} + x\dfrac{dy}{dx} - 2y = 0, \quad y(0) = 0, \quad y'(0) = 1;$

   (c) $(x^2 + 1)\dfrac{d^2y}{dx^2} + x\dfrac{dy}{dx} + 2xy = 0, \quad y(0) = 2 \quad y'(0) = 3;$

   (d) $(2x^2 - 3)\dfrac{d^2y}{dx^2} - 2x\dfrac{dy}{dx} + y = 0, \quad y(0) = -1, \quad y'(0) = 5;$

   (e) $x\dfrac{d^2y}{dx^2} + \dfrac{dy}{dx} + 2y = 0, \quad y(1) = 2, \quad y'(1) = 4.$

3. Identify the $regular\ singular$ point(s) x_0 in the following problems and solve near x_0 using Frobenious method:

   (a) $9x(1 - x)y'' - 12y' + 4y = 0;$

   (b) $xy'' + y - xy = 0;$

   (c) $16x^2y'' + (x^2 + 3) = 0;$

   (d) $x^2y'' + 6xy' + (6 - x^2y) = 0;$

(e) $x^2(x-2)^2y'' + 2(x-2)y' + (x+1)y = 0$;

(f) $2x^2y'' + (x-x^2)y' - y = 0$;

(g) $x^2(x^2-1)y'' - x(1+x^2)y' + (1+x^2)y = 0$;

(h) $xy'' + (1-2x)y' + (x-1)y = 0$;

(i) $x(x-1)y'' + 2(2x-1)y' + 2y = 0$.